

AI OPERATIONS COMMAND CENTER

Managing, Automating and Optimizing Project
Operations Through Conversational AI Agents



TRACK

Track 04 – Stateful AI Agents with Dynamic UI



DOMAIN

AI • Enterprise Productivity • Workflow Automation



PROBLEM STATEMENT

Managing, Automating and Optimizing Project
Operations Through Conversational AI Agents



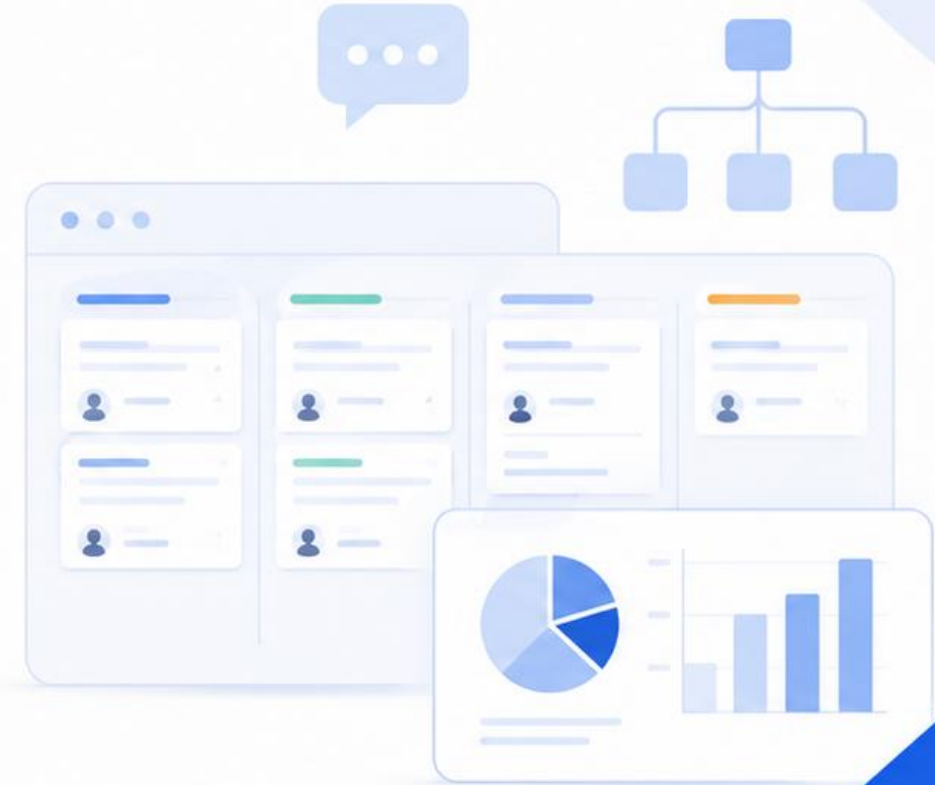
TEAM NAME
CIPHER



COLLEGE NAME
St. Joseph's College
of Engineering



TEAM MEMBERS
• S. Barath Kumar
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CIPHER

IDEA TITLE

ABSTRACT:

AI Operations Manager is a conversational AI platform that enables startups to manage projects using **natural language**. Users can create tasks, assign team members, track bugs, generate sprint plans, and monitor progress through a **single chat interface**. The system **automatically updates** databases, renders live Kanban boards, task tables, dashboards, and analytics, while providing intelligent recommendations for resource allocation and project planning, **reducing manual effort** and improving team productivity.

INNOVATION & UNIQUENESS:

- Chat-Based Project Management
- AI Sprint Planning
- Live Dashboard Generation
- Instant Task Updates
- Smart Team Allocation
- All-in-One Workspace

PROBLEMS

Multiple project management tools

Time wasted updating dashboards

Manual task assignment

Delayed project visibility

Poor sprint planning

Lack of real-time insights

SOLUTIONS

Natural language project management

AI-generated sprint planning

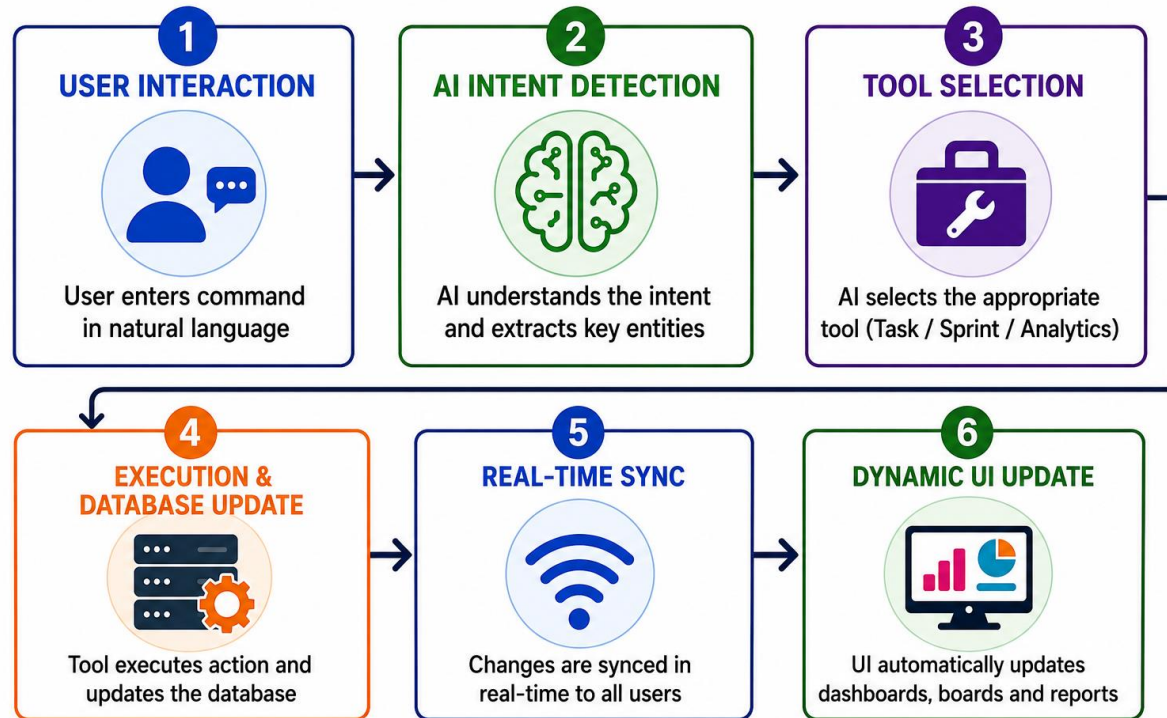
Live Kanban board generation

Real-time task tracking

Instant database updates

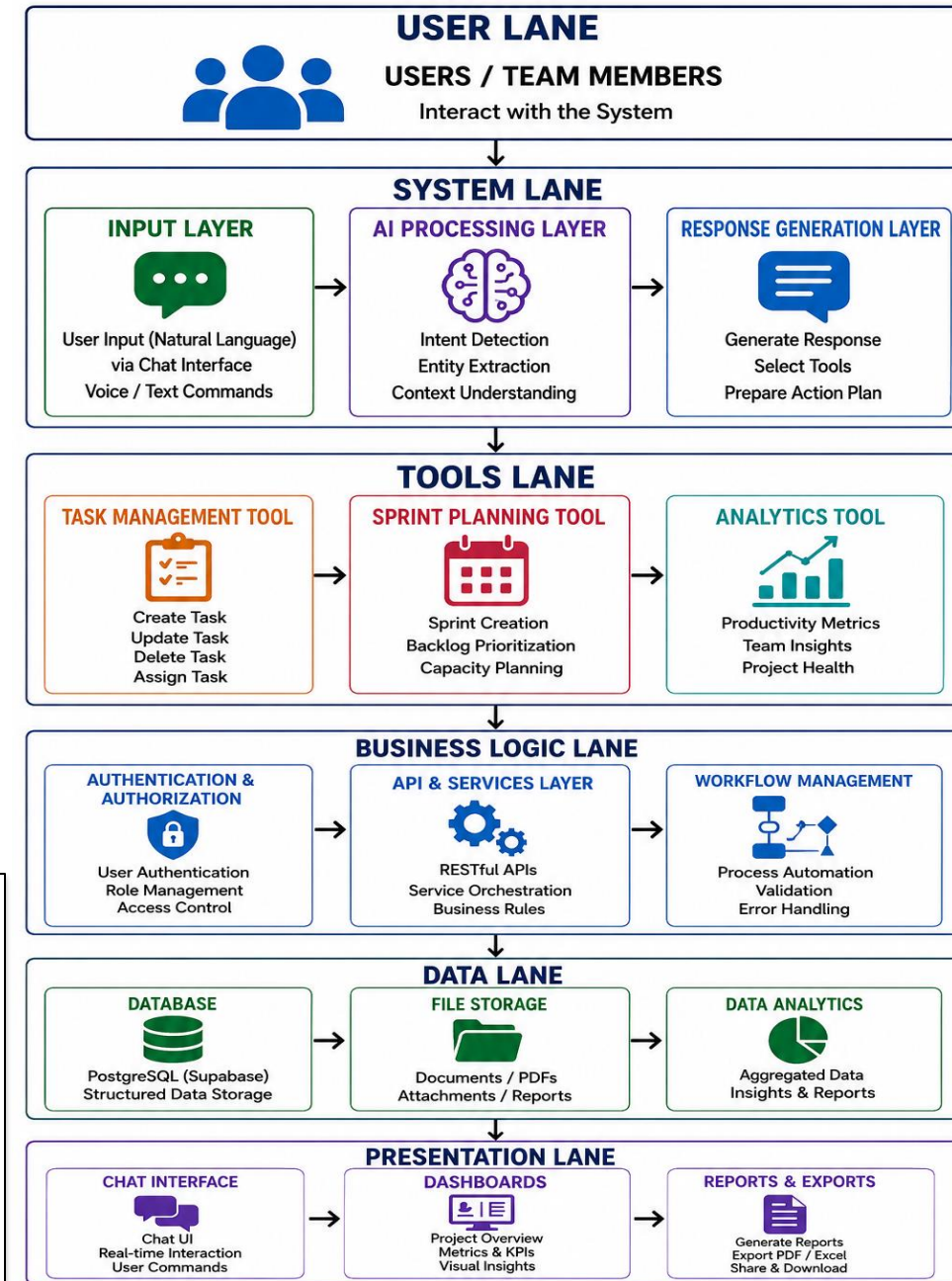
AI-powered project analytics Predictive workload balancing

PROCESS WORKFLOW



TECH STACK

- ❖ **Backend:** Python, FastAPI , AI Engine:, Llama 3 ,OpenAI API,LangChain
- ❖ **Database:** PostgreSQL , SQLite
- ❖ **Frontend:** ReactJS , Tailwind CSS
- ❖ **Authentication:** JWT
- ❖ **Deployment :** Docker
- ❖ **Visualization :** Chart.js



FEASIBILITY AND VIABILITY

CHALLENGES

Intent understanding

Database consistency

Real-time updates

Team collaboration

Large projects



SOLUTIONS

Fine-tuned prompts

Transaction management

WebSockets

Role-based access

Optimized queries

Why Feasible

- Built using proven technologies
- Low infrastructure cost
- Easy deployment
- Scalable architecture
- Supports startups of any size
- Works on cloud or local servers

STRATEGIES

- AI + rule-based validation
- Continuous model improvement
- Scalable microservice architecture
- Automated backups
- Security-first design

IMPACT AND BENEFITS

QUANTIFIABLE IMPACT

- **Task Assignment:**
15 minutes → 5 seconds
- **Sprint Planning:**
2 hours → 2 minutes
- **Bug Tracking:**
Manual → Automated
- **Dashboard Updates:**
30 minutes → Instant
- **Project Visibility:**
Delayed → Real-Time

Economic Benefits

- Reduce project delays
- Improve team productivity
- Lower operational costs
- Faster product delivery

Social Benefits

- Better team collaboration
- Reduced managerial workload
- Improved employee efficiency

Business Benefits

- Real-time visibility
- Better decision making
- Faster sprint execution
- Improved customer satisfaction

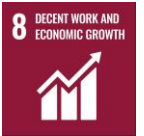
TARGET AUDIENCE

- ✓ Startups
- ✓ SaaS Companies
- ✓ Product Teams
- ✓ Software Development Firms
- ✓ Project Managers
- ✓ Remote Teams

RESEARCH AND REFERENCES

- Agile Project Management Guide [Click here](#)
- Atlassian Jira Documentation [Click here](#)
- Scrum Framework Guide [Click here](#)
- Lang Chain Documentation [Click here](#)
- Fast API Documentation [Click here](#)
- PostgreSQL Documentation [Click here](#)
- Docker Documentation [Click here](#)

SUSTAINABLE DEVELOPMENT GOALS



SDG 8 – Decent Work and Economic Growth



SDG 9 – Industry, Innovation and Infrastructure



SDG 12 – Responsible Production



SDG 17 – Partnerships for Goals

COMPETITOR COMPARISON

Feature	Jira	Trello	AI Operations Manager
Conversational Management	✗	✗	✓
AI Sprint Planning	✗	✗	✓
Live Dashboard Generation	⚠	⚠	✓
Natural Language Commands	✗	✗	✓
Predictive Analytics	✗	✗	✓